

PROBLEM-SOLUTION FIT ANALYSIS

Developer: [Left Blank] | OptiCrop Project Documentation  
Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Aligning User Pain Points with OptiCrop Features

The following analysis illustrates how OptiCrop's modules address specific agricultural challenges:

- Challenge: Crop Mismatch. Farmers often choose crops that do not align with soil properties, leading to low yields.  
OptiCrop Solution: The Crop Recommendation Classifier evaluates soil and climate parameters to recommend the most compatible crop.  
**Opticrop - Problem-Solution Fit**
- Challenge: Crop-to-soil misalignment -> poor harvest yields  
Problem 1: Crop-to-soil misalignment -> poor harvest yields  
Solution: Random Forest classifier recommends optimal crop from 22 types  
Solution: The NPK Vitality Prescription Engine calculates nutrient imbalances and recommends targeted corrections.
- Challenge: NPK over-application -> soil degradation  
Problem 2: NPK over-application -> soil degradation  
Solution: NPK Correction Engine prescribes Urea / SSP / MOP amendments  
Forecast: NPK over-application -> soil degradation
- Challenge: Yield uncertainty -> financial risk  
Problem 3: Yield uncertainty -> financial risk  
Solution: RF Regressor predicts yield (t/ha) with R2 >= 0.92  
OptiCrop Solution: K-Means Clustering groups soil inputs into diagnostic categories, providing simple insights into soil properties.
- Problem 4: No soil health diagnostics  
Solution: K-Means clustering groups soil into 5 health categories